

VBF-T1R1NOFORCE-CALC-01 - Design Calculation Sheet

Inputs

Input parameters			
Item	Value	Unit	Formula / Source
Nominal cell capacity	5	Ah	Chen2020 dump 'Nominal cell capacity [A.h]'
Voltage window	2.5 - 4.2	V	Chen2020 dump lower/upper cut-off; overcharge = 4.2 + 0.5 = 4.7 V (task spec match)
Electrode area	0.1027	m2	0.065 m x 1.58 m (Chen2020 dump height/width)
Positive thickness	7.56e-05	m	Chen2020 dump
Positive porosity / active fraction	0.335 / 0.665	-	Chen2020 dump
Negative thickness	9.5e-05	m	override (r6_c7_params.json), base 85.2e-6
Negative porosity / active fraction	0.25 / 0.75	-	Chen2020 dump
Separator thickness / porosity	12e-6 / 0.47	m / -	Chen2020 dump
Positive CC (Al) thickness / density	16e-6 / 2700	m / kg.m-3	Chen2020 dump
Negative CC (Cu) thickness / density	12e-6 / 8960	m / kg.m-3	Chen2020 dump
c_max positive / negative	63104 / 33133	mol.m-3	Chen2020 dump
Initial lithiation pos / neg	0.270 / 0.901	-	Chen2020 dump (17038/63104, 29866/33133)
Positive particle radius	1.2e-06	m	override (single-crystal NMC811)
Negative particle radius	8e-07	m	override (fine graphite)
Electrolyte conductivity / diffusivity / t+	1.8 / 5e-10 / 0.5	S/m / m2/s / -	overrides (formulation-bridge estimates)
Cooling h / A	25 / 0.2054	W/m2K / m2	overrides (pouch double-face cold plate)

CapacityEnergy

Capacity and energy			
Item	Value	Unit	Formula / Source
1C discharge capacity	5.654	Ah	r6_c7_1c_spme.json:capacity_ah (1C CC 25 C to 2.5 V)
Discharge energy	20.472	Wh	r6_c7_energy.json:energy_wh = integral V x I_1C dt (I_1C = 5 A)
Midpoint voltage	4.1146	V	r6_c7_energy.json:midpoint_voltage_v
DC resistance (contract)	4.029e-05	Ohm	r6_c7_energy.json:dcr_ohm = (V_start - V@10%) / I_1C
Power density (contract)	2376000	W/kg	r6_c7_energy.json:power_density_w_kg = V_OC^2/(4 DCR)/mass
4C CC charge capacity (true)	4.63	Ah	r6_c7_4c.json:capacity_ah x 5.0 (nominal-capacity factor; runner reports duration x C_rate/3600)
4C CC charge acceptance	81.9	% SOC	4.630 / 5.654; duration ~836 s at 20 A (4C x 5 Ah)
4C charge T_max @45 C	318.79	K	r6_c7_4c.json:T_max_K (lumped thermal)
4C anode potential min	+0.00824 / +0.00515 V		r6_c7_4c.json (SPMe) / r6_c7_4c_dfn.json (DFN) anode_potential_v
Overcharge 4.7 V T_max	298.22	K	r6_c7_oc.json:T_max_K
Thermal runaway triggered	false	-	validation/r6_c7_tr.json:triggered

EnergyDensity

Energy density			
Item	Value	Unit	Formula / Source
Discharge energy	20.472	Wh	r6_c7_energy.json:energy_wh
Contract mass (electrolyte excluded)	44.705	g	r6_c7_energy.json:mass_kg = SUM thickness x (1-porosity) x density x area
Gravimetric energy density	457.93	Wh/kg	20.4719 Wh / 0.044705 kg (r6_c7_energy.json:energy_density_wh_kg)
Stack volume	2.1629e-05	m3	r6_c7_energy.json:volume_m3 = thickness 210.6 um x area
Volumetric energy density	946.52	Wh/L	r6_c7_energy.json:energy_density_wh_l
Electrolyte in mass/volume	excluded	-	r6_c7_energy.json:electrolyte_included = false (parameter set lacks density)

Energy density			
Criterion check	457.93 >= 392.61	Wh/kg	PASS, margin +65.3 (log-evaluate R6)

NPMass

N/P and mass			
Item	Value	Unit	Formula / Source
Positive layer mass	0.163994	kg/m2	r6_c7_energy.json:layer_kg_m2.positive_electrode
Negative layer mass	0.118061	kg/m2	r6_c7_energy.json:layer_kg_m2.negative_electrode
Positive CC (Al) mass	0.0432	kg/m2	r6_c7_energy.json:layer_kg_m2.positive_cc
Negative CC (Cu) mass	0.10752	kg/m2	r6_c7_energy.json:layer_kg_m2.negative_cc
Separator mass	0.002525	kg/m2	r6_c7_energy.json:layer_kg_m2.separator
Negative capacity density	63.28	Ah/m2	33133 x 0.75 x 96485/3600 x 95e-6 (c_max x eps x F/3600 x thickness)
Positive capacity density	85.02	Ah/m2	63104 x 0.665 x 96485/3600 x 75.6e-6
N/P ratio (capacity basis)	0.744	-	63.280 / 85.020; anode-limited parameterization (R4 evidence: 1C capacity tracks anode inventory)
Measured 1C capacity	5.654	Ah	r6_c7_1c_spme.json:capacity_ah

Process

Process parameters			
Item	Value	Unit	Formula / Source
Positive areal density	163.99	g/m2	thickness x (1-porosity) x density = layer_kg_m2 x 1000
Negative areal density	118.06	g/m2	same formula, negative layer
Positive compaction density	2.169	g/cm3	areal density / thickness (163.99 g/m2 / 75.6 um)
Negative compaction density	1.243	g/cm3	118.06 g/m2 / 95 um
Electrolyte fill amount	6.74	g/cell	pore volume 5.6193e-6 m3 x 1.2 g/cm3 (literature density) x fill factor 1.0
Formation recommendation	0.1C CC to 4.2 V, 25 C, 2 cycles	-	design recommended value; production-line value requires tuning